Lesson notes · for you

A stack of neurons

University, 1

Materials
A phone with Darsi; Paper for one hand-worked step; The slides from the kit

Learning goals

- One neuron, one cut
- A stack can bend

Activity: In the lab

1. Open the playground
2. Find a shape a line misses

Discussion

- Why is depth not the same as “more”?

Check yourself

- Name the part that bends the stack.

Try it this week
On paper, compute one forward step: two inputs, two weights, a bias, then a ReLU.

Worksheet

A stack of neurons

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Name _____ Class _____ Date _____

1 **Without an activation, a stack of layers becomes…**

◯ One straight cut ◯ A ring ◯ Two separate networks

2 **Adding a hidden layer can separate shapes one neuron cannot.**

◯ True ◯ False

3 **Match the part to its job.**

Weight	• • • •	A bend, so the stack is not one big line
Bias	• • • •	How much an input counts
Activation	• • • •	A shift before the activation

4 **Put a forward step in order.**

☐ Add the bias

☐ Apply the activation

☐ Multiply inputs by weights

✎ **Where did the single neuron fail, in one sentence?**

Check your answers

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1 **One straight cut**

With no bend between them, the layers multiply out to one linear layer, and a linear layer draws one straight cut.

2 **True**

The playground shows a ring that one line misses and a small network learns.

3 **Weight — How much an input counts**
Bias — A shift before the activation
Activation — A bend, so the stack is not one big line

Without an activation, stacked linear layers collapse into one linear layer.

4 **Top to bottom: 2, 3, 1**

The activation is last. It bends the sum.